Lower Limb Neurological Exam — OSCE Checklist

Deakin MD2 · BB6/BB7 · Time: 8 min (+2 min reading)

Tick each step as you perform or verbalise it. **Shaded rows (*)** *are optional or done if indicated. Order:* **Inspection + gait → Tone → Power → Reflexes → Coordination → Sensation**. *Always compare* **left vs right** *and narrate aloud.*

1. Introduction (WIPER)

☐	**W**ash hands / hand hygiene
☐	**I**ntroduce self (name, role)
☐	**P**atient ID: name and DOB
☐	**E**xplain the exam and gain consent
☐	**E**xpose: legs exposed (underwear / shorts), cover groin; maintain dignity, good lighting
☐	Ask about **pain** before touching
☐	Equipment ready: tendon hammer, cotton wool, neurotip (sharps safety, gentle on thin skin), 128 Hz tuning fork

2. Examination

	Step	What to do / say	Root / abnormal
Inspection + gait (patient standing first)			
☐	General	Walking aids, orthoses, footwear wear, scars (back, legs), skin	Deformity, pes cavus
☐	Muscle bulk	Both sides: quadriceps, calves, tibialis anterior, small foot muscles	Wasting = LMN; symmetry
☐	Fasciculations	Watch thighs and calves at rest	LMN (e.g. MND)
☐	Gait	Walk to wall and back: arm swing, stride, base, turning	Hemiplegic, shuffling, ataxic, high-stepping
☐	Heel walk	"Walk on your heels"	Can't = L4/L5, foot drop
☐	Toe walk	"Walk on your tiptoes"	Can't = S1
☐	Tandem walk	"Heel to toe in a straight line"	Ataxia (cerebellar)
☐	Romberg's	Feet together, arms by side, eyes open then closed; stand close to catch	Sway with eyes closed only = sensory ataxia
Tone (patient lying on bed)			
☐	Leg roll	Roll the leg side to side; foot should flop	Stiff = increased tone
☐	Knee lift	Quickly lift behind knee	Heel lifts off bed = increased tone
☐	Knee + ankle	Passively flex/extend at variable speed	Spasticity / rigidity / hypotonia
☐	Ankle clonus	Knee slightly bent; sharply dorsiflex and hold	≥ 3–5 beats = UMN
Power (MRC 0–5) — isolate the joint, match the patient's strength			
☐	Hip flexion	"Keep your leg straight, lift it up, don't let me push it down"	L2, L3 — iliopsoas
☐	Hip extension	"Push my hand down into the bed" (hand behind calf)	L5, S1, S2 — gluteus maximus
☐	Hip abduction	"Push your legs out against my hands"	L4, L5, S1 — gluteus medius/minimus
☐	Hip adduction	"Squeeze your legs together"	L2, L3, L4 — adductors
☐	Knee flexion	Knee bent: "Pull your heel towards your bottom"	L5, S1 — hamstrings
☐	Knee extension	Knee bent: "Push me away"	L3, L4 — quadriceps
☐	Ankle dorsiflexion	"Pull your foot up, don't let me push it down"	L4, L5 — tibialis anterior
☐	Ankle plantarflexion	"Push down against my hand" (like a pedal)	S1, S2 — gastrocnemius, soleus
☐	Inversion	Hold foot turned in against resistance	L5, S1 — tibialis posterior
☐	Eversion	Hold foot turned out against resistance	L5, S1 — peronei
☐	Great toe extension	"Push your toe up against my finger"	L5, S1 — EHL
☐	Great toe flexion*	"Push your toe down against my finger"	S1, S2 — FHL / FDL
Reflexes — relaxed, positioned; reinforce (clench teeth, Jendrassik) if absent			
☐	Knee jerk	Support both knees slightly flexed; strike patellar tendon	L3, L4
☐	Ankle jerk	Hip externally rotated, knee bent, dorsiflex foot slightly; strike Achilles	S1, S2
☐	Plantar (Babinski)	Blunt object up lateral sole then across; report **first** movement of big toe	Upgoing = UMN
Coordination			
☐	Heel–shin	"Heel on opposite knee, slide down the shin, lift, repeat"	Ataxia, intention tremor
☐	Toe–finger*	"Touch my finger with your big toe"	Past-pointing
☐	Foot tapping	"Tap your foot quickly on my hand"	Dysdiadochokinesia
Sensation — demonstrate on sternum first, eyes closed, compare sides			
☐	Light touch	Cotton wool: dab each dermatome L2–S1	Dermatomal / stocking pattern
☐	Pain	Neurotip: sharp or blunt? Same dermatomes	Spinothalamic loss

	Step	What to do / say	Root / abnormal
☐	Temperature*	Cold tuning fork if indicated	Spinothalamic
☐	Vibration	128 Hz fork on great toe tip; if absent move to MTP → medial malleolus → tibial tuberosity	Dorsal column loss
☐	Proprioception	Hold great toe by its sides; "Up or down?"; if absent move proximally	Dorsal column / sensory ataxia

3. Closing and presenting

☐	Thank patient, help them dress, wash hands
☐	Completion statement: "To complete, I would examine the upper limbs and cranial nerves, examine the spine, and check saddle sensation and anal tone if indicated."
☐	Present one finding per domain: "Gait normal and Romberg's negative; normal bulk; tone normal with no clonus; power 5/5 throughout; reflexes normal, plantars downgoing; coordination normal; sensation intact. Overall a normal lower limb neurological exam."

4. Quick reference

MRC	Power grade
0	No contraction
1	Flicker, no movement
2	Moves with gravity eliminated
3	Moves against gravity, not resistance
4	Against resistance, weaker than normal
5	Normal power

Dermatome	Test site
L2	Anterior upper thigh
L3	Medial knee
L4	Medial leg / medial malleolus
L5	Dorsum of foot, great toe
S1	Lateral foot, little toe, heel
Reflexes	0 absent · 1+ reduced · 2+ normal · 3+ brisk · 4+ clonus

Sign	UMN	LMN
Bulk	Normal / disuse	Wasting
Fasciculations	No	Yes
Tone	Increased, clonus	Decreased
Power	Pyramidal weakness (flexors)	Weak in root/nerve distribution
Reflexes	Brisk, plantar upgoing	Reduced / absent, plantar down

Gait	Think of
Hemiplegic	Circumduction — stroke
Shuffling	Small steps, reduced arm swing — Parkinson's
Broad-based ataxic	Cerebellar lesion
High-stepping	Foot drop — common peroneal / L5
Waddling	Proximal myopathy

- **Pyramidal pattern (UMN):** flexors weaker than extensors in the leg — hip flexion, knee flexion and dorsiflexion go first.
- **Foot drop:** common peroneal = weak dorsiflexion + eversion, **inversion spared**; L5 root = inversion also weak.
- **Romberg's positive** = proprioceptive (dorsal column) loss, not cerebellar. **Stocking** loss = peripheral neuropathy.
- **Red flags:** saddle anaesthesia, urinary retention, bilateral leg weakness = cauda equina — urgent MRI.
- **Pitfalls:** not offering to catch in Romberg's, overpowering the patient, skipping reinforcement, testing sensation with eyes open.